Pei Zhang

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Last updated: August 19, 2026

Education

2019/09 - 2025/05 **Ph.D. in Mathematical Statistics, University of Maryland, College Park** (UMD)
Dissertation title: *Mixed Modeling Approaches for Characterizing Genetic Effects and Heritability Metrics in Longitudinal Phenotypes* (NIH Outstanding Graduate Research)
Advisors: *Drs. Paul S. Albert (NIH Biostatistics), Doron Levy (UMD Applied Mathematics), Jianxin Shi (NIH Biostatistics) and Hyokyoung G. Hong (NIH Biostatistics)*

2014/09 - 2016/06 **M.S. in Mathematics, University of Science and Technology of China** (USTC)
Dissertation title: *Ekeland Index Theory for Periodic Solutions and Symmetric Periodic Solutions of Convex Hamiltonian Systems*
Advisors: *Drs. Xinan Ma and Hui Liu*

2010/09 - 2014/06 **B.S. in Mathematics and Applied Mathematics (Teacher Education Program), Nanjing Normal University** (NNU)
Dissertation title: *Some Problems in Hamiltonian Systems* (NNU Outstanding Graduation Thesis)
Advisor: *Dr. Yujun Dong*

2009/09 - 2010/06 **Undergraduate in Computer Science, NNU**

Employment History

2025/05 - present **Postdoctoral Fellow (Mentored by Drs. Paul S. Albert and Jianxin Shi)**
Biostatistics Branch (BB), Division of Cancer Epidemiology & Genetics (DCEG)
National Cancer Institute (NCI), National Institutes of Health (NIH)

2021/09 - 2025/05 **Predoctoral Fellow (Mentored by Drs. Paul S. Albert and Jianxin Shi, and Hyokyoung G. Hong)**
BB, DCEG
NCI, NIH

2019/08 - 2021/08 **Teaching Assistant**
Department of Mathematics, UMD

2016/08 - 2017/04 **Mathematics Teacher**
NO.1 Middle School Affiliated to Central China Normal University, Wuhan, China

2014/09 - 2016/06 **Teaching Assistant**
Department of Mathematics, USTC

Research Interests

- **Longitudinal and correlated data methodology**, with emphasis on repeated-measures analysis, biomarker studies, and complex dependent data structures.
- **Statistical and computational methods in genetics** for analyzing electronic health records (EHR), real-world data (RWD), and large-scale population studies (e.g., the Prostate, Lung, Colorectal, and Ovarian (PLCO) Cancer Screening Trial), with applications to prostate cancer risk stratification, early detection, and prediction.
- **Statistical and machine learning approaches in epidemiology**, including machine learning and deep learning for cancer genetic epidemiology, nutritional epidemiology, and metabolomics.

Publications

* corresponding author † co-first / equal contribution ‡ alphabetical order

Methodological / Statistical Papers

1. Bandreddi, Sumaja†, **Zhang, Pei**†*, R. Kelly, M. Machiela, and Albert, Paul S.*, "Analysis of semi-continuous longitudinal data using tobit and two-part hurdle models with application to clonal hematopoiesis," under review at *Statistics in Medicine*, 2026.
2. **Zhang, Pei**, X. Wang, Shi, Jianxin*, and Albert, Paul S.*, "Estimating the heritability of longitudinal rate-of-change: Genetic insights into PSA velocity in prostate cancer-free individuals," *Biostatistics*, vol. 27, no. 1, kxag015, 2026, [PDF].
3. **Zhang, Pei**, Albert, Paul S.*, and Hong, Hyokyoung G.*, "Mixed modeling approach for characterizing the genetic effects in a longitudinal phenotype," *Annals of Applied Statistics*, vol. 19, no. 3, pp. 2070–2087, 2025, [PDF].

Collaborative / Applied Papers

1. H. Fan, H. Zhao, **Zhang, Pei**, P. Yu, Y. Ji, G. Chen, H. Jin, Y. Liu, J. Liu, Z.-S. Chen, Lyu, Aiping*, Liang, Xinmiao*, and Chen, Yang*, "Homoisoflavanone delays colorectal cancer progression via DNA damage-induced mitochondrial apoptosis and parthanatos-like cell death," *Advanced Science*, e11406, 2026, [PDF].
2. H. Fan, H. Zhao, L. Gao, Y. Dong, **Zhang, Pei**, P. Yu, Y. Ji, Z.-S. Chen, X. Liang, and Chen, Yang*, "CCN1 enhances tumor immunosuppression through collagen-mediated chemokine secretion in pancreatic cancer," *Advanced Science*, e2500589, 2025, [PDF].
3. Loftfield, Erikka†*, **Zhang, Pei**†, C. P. O'Connell, L. L. Kahle, K. Herrick, L. Abar, N. Khandpur, E. M. Steele, and H. G. Hong, "Performance of a food frequency questionnaire for estimating ultraprocessed food intake according to the NOVA classification system in the United States NIH-AARP diet and health study," *Journal of Nutrition*, vol. 155, no. 7, pp. 2376–2384, 2025, [PDF].
4. Wu, Yetian†, **Zhang, Pei**†, H. Fan, C. Zhang, P. Yu, X. Liang, and Chen, Yang*, "GPR35 acts a dual role and therapeutic target in inflammation," *Frontiers in Immunology*, vol. 14, no. 1254446, 2023, [PDF].

Preprints / Under Review

1. Lim, Jungeun, **Zhang, Pei**, J. Rhee, M. C. Playdon, A. J. Cross, R. Stolzenberg-Solomon, V. L. Roger, M. P. Purdue, Albanes, Demetrius*, Hong, Hyokyoung G.*, and Wong, Jason Y. Y.*, "Machine learning-derived metabolomic profile scores predict serum PFOS and PFOA levels in a US cohort," under review with minor revision at *Scientific Reports*, 2026.

2. M. Zhang, X. Zhang, L. Wang, X. Liu, X. Zhang, J. Li, K. Wang, H. Li, H. Koka, T. Ma, W. Luo, S. Chavez, Y. Kim, A. Morales Miranda, T. Veith, **Zhang, Pei**, W. Zhou, T. Zhang, X. Yang, Y. Zhang, T. Zhang, and J. Bai, "A single-cell atlas of chordoma reveals a primitive malignant state linked to immune remodeling and clinical risk," under review at *Cancer Communications*, 2026.

PhD Thesis

1. **Zhang, Pei**, "Mixed modeling approaches for characterizing genetic effects and heritability metrics in longitudinal phenotypes," [PDF], Ph.D. dissertation, University of Maryland, College Park, 2025.

Honors

2026	NSF-Supported Travel Award, NISS Writing Workshop
	Finalist (one of six selected finalists) for the IBS/CWS Florence Nightingale Award, International Biometric Conference 2026, Seoul, South Korea
	Certificate of Completion of Easylearning NLP GenAI Agent Course
	National Science Foundation (NSF) Travel Support Award, USA
	Institute of Mathematical Statistics (IMS) New Researcher Travel Award, USA
2025 & 2026	NIH Postdoctoral Fellowship (twice)
2025	NCI-DCEG Institutional Nominee for NCI Pathway to Independence Award for Outstanding Early Stage Postdoctoral Researchers (K99/R00) [about "Early K99" Grant - NIH] | [about "Early K99" Grant - Stanford Medicine] | [Notice of Funding Opportunity Number: PAR-23-286]
	NIH Fellows Award for Research Excellence (FARE) 2026 (top 25% of all applicants) [List of FARE 2026 Awardees Abstracts] | [about FARE]
	NIH Outstanding Graduate Research
2023 & 2024	Women in Statistics and Data Science Conference Student Travel Award (twice)
2023	UMD Jacob K. Goldhaber Travel Grant
	UMD International Conference Student Support Award
2022 & 2023	UMD Graduate Student Travel Fellowship (twice)
2022	Rutgers University Travel Award for Conference on Advances in Bayesian and Frequentist Statistics
2021 - 2024	UMD Tuition Award for External Fellowship (4 times)
	NIH Predoctoral Fellowship (4 times) [about NIH Graduate Partnerships Program (GPP)]
2019 & 2020	UMD Dean's Fellowship (twice)
2017	First Award in Thesis Contest on Deep Integration of Key Competence and Classroom Teaching (NO.1 Middle School Affiliated to Central China Normal University)
2016	USTC Outstanding Graduate
2014 & 2015	USTC First Award of Academic Scholarship (twice)
2014	NNU Outstanding Graduate
	NNU First Award of Feng Ruer Scholarship
	NNU Outstanding Dissertation
2013	Certificate of Recommendation for Exemption from Entrance Examination—USTC Graduate Program
	Eligibility for Recommended Admission (top 5% in Mathematics Teacher Education Program, NNU)

Honors (continued)

- Second Award of Jiangsu Teaching Professional Skills Contest (ranked 4th in Mathematics across the Jiangsu Province)

2012
- NNU Second Award in Mathematical Modeling Contest
- NNU Outstanding Debate Team Award, Debate Competition on "Is Beauty Subjective or Objective?" Member of the award-winning team arguing that beauty is objective.

2010 & 2011
- NNU First Award of Rolling Scholarship (twice)

2009
- NNU First Award in Computer Contest

Academic Activities

Invited Presentations

2026/07
- International Biometric Conference (IBC), Seoul, South Korea (Invited Talk)

2026/05
- Biostatistics Seminar, UC Davis Comprehensive Cancer Center, Davis, California, USA (Invited Talk)
- STATGEN 2026: Conference on Statistics in Genomics and Genetics, Emory University, Atlanta, Georgia, USA (Invited Poster)
- The 10th Annual NIH Vivian Pinn Symposium, Bethesda, Maryland, USA (Invited Poster)

2026/01
- Biostatistics Branch of NCI-DCEG Works in Progress, Rockville, Maryland, USA (Invited Talk)

2024/10
- Women in Statistics and Data Science (WSDS), Reston, Virginia, USA (Invited Panelist)

Contributed Presentations

2026
- The American Society of Human Genetics (ASHG) Annual Meeting, Montreal, Canada
- Joint Statistical Meetings (JSM), Boston, Massachusetts, USA
- The 11th Workshop on Biostatistics and Bioinformatics, Georgia State University, Atlanta, Georgia, USA (Contributed Poster as a recipient of the IMS New Researchers Travel Award and NSF Travel Support Award)
- AI Day for Federal Statistics, National Academy of Sciences, Washington DC, USA

2025
- Joint Statistical Meetings (JSM), Nashville, Tennessee, USA
- The 17th Annual NCI-DCEG Fellows' Training Symposium, Rockville, Maryland, USA

2024
- Joint Statistical Meetings (JSM), Portland, Oregon, USA
- AI Day for Federal Statistics, National Academy of Sciences, Washington DC, USA
- Probability and Statistics Day, University of Maryland, Baltimore County (UMBC), Baltimore, Maryland, USA
- NIH Annual Graduate Student Research Symposium, Bethesda, Maryland, USA

2023
- Women in Statistics and Data Science (WSDS), Seattle, Washington, USA
- Joint Statistical Meetings (JSM), Toronto, Canada
- Probability and Statistics Day, University of Maryland, Baltimore County (UMBC), Baltimore, Maryland, USA

2022
- Conference on Advances in Bayesian and Frequentist Statistics with a Celebration of the 80th Birthday of Professor William E. Strawderman, Rutgers University, New Brunswick, New Jersey, USA

Academic Activities (continued)

- Annual Conference on Quantitative Approaches in Biology, Northwestern University, Chicago, Illinois, USA

Attendant

2026
- The 31st CBA Annual Conference, Gaithersburg, Maryland, USA
- Artificial Intelligence (AI) Expo, The Department of Health and Human Services (HHS), Washington D.C., USA
- NIH Library Artificial Intelligence (AI) Trainings led by OpenAI experts, Bethesda, Maryland, USA
- Biostatistics Symposium of Southern California (BSSC), Newport Beach, California, USA

2025
- Eastern North American Region (ENAR), New Orleans, Louisiana, USA

2024
- Statistics in the Age of AI Conference, The George Washington University (GWU), Washington DC, USA
- Eastern North American Region (ENAR), Baltimore, Maryland, USA

2023
- National Heart, Lung, and Blood Institute (NHLBI) Biostatistics Workshop on Clinical Trial Designs: Innovative Endpoints and Futility Monitoring, Bethesda, Maryland, USA

2022
- International Small Area Estimation (SAE) Annual Conference, UMD, College Park, Maryland, USA

2016
- National Tianyuan Foundation Summer School on Pure Mathematics, Summer Seminar on Pure Mathematics, Zhejiang University, Hangzhou, China

2013
- Yangtze River Delta Symposium on Partial Differential Equations and Doctoral Student Forum, Tongji University, Shanghai, China

Grant Writing Workshop

2026
- NCI K Grant Writing Workshop, Bethesda, Maryland, USA
- NCI-DCEG Writing Your Specific Aims Session, Rockville, Maryland, USA

2025
- NCI-DCEG Grant Writing Workshop, Rockville, Maryland, USA

Software

R Packages and Research Code

- **Heritability-Velocity** — Author and Maintainer. Reproducible framework for joint estimation of static and dynamic heritability components in longitudinal phenotypes, featuring scalable AI-REML and REHE algorithms for genome-wide data. [GitHub] | [arXiv]

- **PLCO** — Author and Maintainer. Mixed-effects modeling framework with crossed genetic and individual-specific random effects to quantify genetic influences on PSA trajectories in large-scale prostate cancer screening studies. [GitHub] | [Paper]

- **AARP** — Author and Maintainer. Implementation of measurement error models for bias correction in food frequency questionnaire–based dietary intake assessment. [GitHub] | [Paper]

Software (continued)

Programming and Statistical Computing

- R | Python | C/C++ | SQL | Jupyter Notebook

Research Infrastructure

- NIH Biowulf High-Performance Computing Cluster | All of Us Research Hub | UK Biobank

Scientific Writing and Reproducibility

- LaTeX | Microsoft Word | R Markdown | Git

Teaching

Teaching Assistant / Grader

Fall 2019	**Sampling Theory (STAT440)**, Undergraduate/master-level. *Grader, UMD*
Spring 2020	**Introduction to Linear Algebra (MATH240)**, Undergraduate-level. *Teaching Assistant, UMD*
Summer 2020	**Introduction to Mathematical Proof (MATH310)**, Undergraduate-level. *Grader, UMD*
	Introduction to Probability Theory (STAT410), Undergraduate/master-level. *Grader, UMD*
Fall 2020 & Spring 2021	**Elementary Probability and Statistics (STAT100)**, Undergraduate-level. *Teaching Assistant, UMD*
Spring 2015 & 2016	**Calculus II**, Undergraduate-level. *Teaching Assistant, USTC*
Fall 2016	**Calculus I**, Undergraduate-level. *Teaching Assistant, USTC*

Mathematics Teacher (Internship)

2013/09 - 2013/11	**High School Mathematics**. High school level. *Mathematics Teacher Intern, Jinling High School, Nanjing, China*

Service

Journal Reviewer

2021 - present	Biometrics (1); Statistics in Medicine (2); BMC Medical Research Methodology (1); Journal of Biopharmaceutical Statistics (1); Frontiers in Oncology (1); LabMed Discovery (1)

Judging

2026	Chief Judge, NIH Fellows Award for Research Excellence (FARE) 2027 Competition

Service (continued)

Conference / Seminar Organizer

2024
- Co-organizer of the panel "Discussion of Mentoring Program and Participant Experiences" at WSDS 2024
- Chair of the session "Rethinking Current Modeling Strategies: Advances and Applications" at JSM 2024

2023
- Chair of the session "Concurrent - Statistical Insights into Mortality and Inference: From Choquet Capacities to Demographic Patterns" at WSDS 2023

Mentoring

2025–2026
- Co-mentor (with **Paul S. Albert, PhD**) of postbaccalaureate fellow **Sumaja Bandreddi** (M.S., B.A., University of Virginia), Biostatistics Branch, DCEG, NCI, NIH

Spring 2021
- Directed Reading Program mentor for undergraduate **Andrew Craun**, Department of Mathematics, University of Maryland, College Park

Leadership

2026
- Member, NIH Fellows Award for Research Excellence (FARE) 2027 Committee

2022–2023
- Vice President, Women in Mathematics, University of Maryland, College Park

2020–2021
- Representative, Graduate Student Government, University of Maryland, College Park

2010–2011
- Team Leader, Volunteer Interpreters Group, Nanjing Yunjin Museum, Nanjing, China

Volunteer Service

2026
- Volunteer, The Chinese Biopharmaceutical Association-USA (CBA)

2025
- Volunteer, Sino-American Pharmaceutical Professionals Association (SAPA)-DC

2022
- Volunteer Interviewee, The Center for Mathematics Education, University of Maryland, College Park

2009–2010
- Volunteer Teacher, Nanjing Hongshan Primary School for Children of Migrant Workers, Nanjing, China

Memberships

2026 - present
- American Society of Human Genetics

2024 - present
- Washington Statistical Society
- International Chinese Statistical Association
- Section of Statistics in Genetics and Genomics

2023 - present
- American Statistical Association
- Institute of Mathematical Statistics
- Society for Mathematical Biology
- Caucus for Women in Statistics and Data Science
- Eastern North America Region/International Biometric Society

2019 - 2025
- American Mathematical Society

References

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Doron Levy
Professor of Mathematics
Chair, Department of Mathematics
Director, Brin Mathematics Research Center
Co-director, the NCI-UMD Partnership for Integrative Cancer Research
Fellow of the Society for Industrial and Applied Mathematics (SIAM)
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